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Inventor(s)	Lefebvre; Marie et al.

Weaving loom with tiltable reed and weaving method

Abstract

A Jacquard type loom intended to produce a fibrous structure by three-dimensional weaving between a plurality of warp yarns and a plurality of weft yarns extending in a horizontal direction, the loom including a reed movable between a rest position upstream of the rapier and a beat-up position of a fell of the fibrous structure in a beat-up direction parallel to the horizontal direction. The reed includes at least two stationary portions forming an angle therebetween. The reed is mounted on the loom by a positioning mechanism able to orient the reed about an axis of rotation so as to adjust an angular position of the stationary portions of the reed relative to a horizontal reference plane. The loom include vertical translation elements able to move the reed or the fell of the fibrous structure in the vertical direction.

Inventors:	Lefebvre; Marie (Moissy-Cramayel, FR), Planckeel; Aline (Moissy-Cramayel, FR), Coupe; Dominique Marie Christian (Moissy-Cramayel, FR)
Applicant:	SAFRAN CERAMICS (Le Haillan, FR); SAFRAN (Paris, FR)
Family ID:	1000008617508
Assignee:	SAFRAN CERAMICS (Le Haillan, FR); SAFRAN (Paris, FR)
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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) This application is the U.S. National Stage of PCT/FR2023/050854, filed Jun. 13, 2023, which in turn claims priority to French patent application number 22 06080 filed Jun. 21, 2022. The content of these applications are incorporated herein by reference in their entireties.

TECHNICAL FIELD

(2) The present invention relates to a Jacquard type loom and to the production of composite material parts and more particularly the production by three-dimensional (3D) weaving of fibrous reinforcement structures for such parts.

PRIOR ART

(3) A field of application of the invention is the production of parts made of structural composite material, i.e. structural parts with fibrous reinforcement densified by a matrix, such as parts made of organic matrix composite material (OMC), carbon matrix composite material (C/C) and ceramic matrix composite material (CMC). The organic matrix composite material (OMC), carbon matrix composite material (C/C) and ceramic matrix composite material (CMC) replace the metal material parts in some portions of the turbomachines. Their use contributes to optimizing the performance of the aircrafts, in particular by improving the efficiency of the turbomachine and decreasing the overall mass of the turbomachine, which significantly reduces harmful emissions to the environment (CO, CO.sub.2, NO_x, etc.).

(4) The invention more particularly relates to monitoring the direction of juxtaposition of the weft yarns in the columns of weft yarns during the weaving.

(5) FIG. 1 schematically illustrates a Jacquard type loom **310** used for the production of fibrous textures or fabric obtained by multilayer weaving between a plurality of layers of warp yarns **330** and a plurality of layers of weft yarns **331**.

(6) In a known manner, the loom **310** is equipped with a Jacquard mechanism **3311** supported by a superstructure not represented in FIG. 1. The loom **10** also comprises a harness **320** consisting of a tying-up board **321** and of control yarns or heddles **322**, each heddle **322** being connected at one end to a control hook **12** of the Jacquard mechanism **311** and at the other end to one of the return springs **313** fixed to the frame **314** of the loom **310**.

(7) Each heddle **322** comprises an eyelet **323** crossed by a warp yarn **330**. The heddles **322** and their associated eyelet **323** are driven by a substantially vertical oscillating movement represented by the double arrow F under the tensile forces exerted respectively by the control hooks **312** and the return springs **313**. The heddles **322** make it possible to lift certain warp yarns **330** and thus create a shed **315**, that is to say an opening between lower and upper webs of warp yarns **330** allowing the introduction of weft yarns **331** by means of a rapier **341**. The rapier **341** is equipped at its end with a clamp **342** that grips a weft yarn **331** from a bobbin **340** to place it through the opening between the webs of warp yarns **330** created by the shed **315**, the weft yarn **331** being cut using a knife **343** after its positioning in the shed **315**.

(8) A reed **350** present upstream of the rapier **341** in its rest position is then folded down in order to pack the weft yarn(s) introduced into the shed **315**. The rapier **341** is then ready to again grip a new weft yarn **331** from the bobbin **40** and place it. A fibrous structure is thus gradually formed having a three-dimensional (3D) weaving between the warp yarns **330** and the weft yarns **31**.

(9) If this type of loom makes it possible to produce complex, i.e. 3D, weaving, all of the columns of weft yarns are oriented at 90° relative to a horizontal direction D.sub.H. However, there is an advantage in being able to position the weft columns at an angle other than 90°, particularly to pre-

position the wefts before shaping the fibrous structure or to define specific mechanical properties in the final part.

DISCLOSURE OF THE INVENTION

(10) For this purpose, the present invention proposes a Jacquard type loom intended to produce a fibrous structure by three-dimensional weaving between a plurality of warp yarns and a plurality of weft yarns extending in a horizontal direction, the loom comprising a plurality of control yarns able to move the warp yarns in a vertical direction, at least one rapier present downstream of the control yarns able to pull a weft yarn and a reed movable between a rest position upstream of the rapier and a beat-up position of a fell of the fibrous structure in a beat-up direction parallel to the horizontal direction, characterized in that the reed comprises at least two stationary portions forming an angle therebetween and in that the reed is mounted on the loom by means of a positioning mechanism able to at least orient the reed about an axis of rotation so as to adjust an angular position of said at least two stationary portions of the reed relative to a horizontal reference plane, the loom further comprising vertical translation means able to move the reed or the fell of the fibrous structure in the vertical direction so as to move said reed or said fell upwards or downwards relative to the horizontal reference plane.

(11) In the loom of the invention, the angle with which the reed beats up the fell can be adjusted to a value other than 90° relative to a horizontal reference plane. It is thus possible to weave columns of weft yarns in which the weft yarns are juxtaposed in a layering direction not perpendicular to the horizontal reference plane. The inclination of the layering direction is monitored by the choice of the value of the beat-up angle of the reed.

(12) According to one particular characteristic of the loom of the invention, the angle formed between said at least two stationary portions is comprised between 60° and 80° , more preferably an angle of about 70° .

(13) The invention also relates to a method for three-dimensional weaving in a single piece of a fibrous structure between a plurality of layers of warp yarns and a plurality of layers of weft yarns, the weaving being performed by a loom according to the invention, the warp yarns extending in a horizontal direction corresponding to the direction of advance of said warp yarns, the weft yarns extending in a transverse direction, the weft yarns being woven in a plurality of columns spaced from each other in the horizontal direction, each column of weft yarns being positioned by the reed of the loom against the fell of the fibrous structure, the weft yarns of each column of weft yarns being juxtaposed in the thickness of the fibrous structure in a determined layering direction, the method comprising the weaving of at least one section with inclined weft columns in the fibrous structure, the weft yarns of the columns of weft yarns of said at least one section with inclined weft columns being juxtaposed in at least one first layering direction not perpendicular to a horizontal reference plane.

(14) A fibrous structure is thus formed comprising one or more deployment sections in which the weft yarns in the columns of weft yarns are juxtaposed in a layering direction forming an angle with the horizontal direction other than 90° .

(15) According to one particular characteristic of the method of the invention, the weft yarns of the column of weft yarns of said at least one section with inclined weft columns are juxtaposed in different first and second layering directions.

(16) A section of fibrous structure is thus formed with columns of weft yarns having a broken shape.

(17) According to another particular characteristic of the method of the invention, said at least one section with inclined weft columns is interlinked by weaving to one or more adjacent sections of the fibrous structure, the weft yarns of the columns of weft yarns of the adjacent section(s) being juxtaposed in a layering direction perpendicular to the horizontal direction.

(18) The section(s) with inclined weft columns may here correspond to one or more deployment sections which are folded during the shaping of the fibrous structure.

(19) Thus, after the folding of the section(s) with inclined weft columns, the angular variation resulting from the shear forces exerted therein during their deployment causes the horizontal straightening of the weft yarn columns in these sections and prevents detachment in the fibrous structure at the level of the inner radius.

(20) According to another particular characteristic of the method of the invention, the first layering direction in said at least one section with inclined weft columns forms an angle with the horizontal direction comprised between 60° and 80° , more preferably an angle of about 70° .

(21) The invention also relates to a method for manufacturing a composite material part comprising: weaving a fibrous structure in accordance with the weaving method according to the invention, shaping the fibrous structure so as to obtain a fibrous preform, densifying the fibrous preform by a matrix.

(22) The invention further relates to a fibrous structure having a three-dimensional weaving between a plurality of layers of warp yarns and a plurality of layers of weft yarns, the warp yarns extending in a horizontal direction, the weft yarns extending in a transverse direction, the structure comprising a plurality of columns of weft yarns spaced from each other in the horizontal direction, the weft yarns of each column of weft yarns being juxtaposed in the thickness of the fibrous structure in a determined layering direction, the fibrous structure comprising at least one section with inclined weft columns, the weft yarns of the columns of weft yarns of said at least one section with inclined weft columns being juxtaposed in at least one first layering direction not perpendicular to the horizontal direction.

(23) According to one particular characteristic of the fibrous structure of the invention, the weft yarns of the columns of weft yarns of said at least one section with inclined weft columns are juxtaposed in different first and second layering directions. The fibrous structure thus includes one or more sections with weft yarn columns having a broken shape.

(24) According to another particular characteristic of the fibrous structure of the invention, said at least one section with inclined weft columns is interlinked by weaving to one or more adjacent sections of the fibrous structure, the weft yarns of the columns of weft yarns of the adjacent section(s) being juxtaposed in a layering direction perpendicular to the horizontal direction.

(25) According to another particular characteristic of the fibrous structure of the invention, the first layering direction in said at least one section with inclined weft columns forms an angle with the horizontal direction comprised between 60° and 80° , more preferably an angle of about 70° .

(26) The invention also relates to a composite material part comprising a fibrous reinforcement densified by a matrix, characterized in that the fibrous reinforcement comprises a fibrous structure according to the invention.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1 is a schematic perspective view of a Jacquard-type loom according to the prior art,

(2) FIG. 2 is a schematic perspective view of a Jacquard-type loom according to one embodiment of the invention,

(3) FIG. 3 is a side view of the loom of FIG. 2 showing the weaving of a section of fibrous structure with weft yarn columns perpendicular to a horizontal reference plane,

(4) FIG. 4 is a side view of the loom of FIG. 2 showing the weaving of a section of fibrous structure with inclined weft yarn columns in a layering direction,

(5) FIG. 5 is a side view of the loom of FIG. 2 showing the weaving of a section of fibrous structure with inclined weft yarn columns in two different layering directions,

(6) FIG. 6 is a side view of the loom of FIG. 2 showing the weaving of a section of fibrous structure with inclined weft yarn columns in two different layering directions.

DESCRIPTION OF THE EMBODIMENTS

(7) The invention generally applies to Jacquard type looms used in particular to produce fibrous structures or fabrics by three-dimensional (3D) weaving between layers of warp yarns and layers of weft yarns. The invention applies more particularly in the case of 3D weaving of fibrous structure with at least one section with inclined weft columns as explained in detail below. By “three-dimensional weaving” or “3D weaving” is meant here a weaving mode by which at least some of the weft yarns interlink warp yarns on several layers of warp yarns or vice versa. One example of three-dimensional weaving is called “interlock” weave. By “interlock” weaving is meant here a weave in which each layer of warp yarns interlinks several layers of weft yarns, with all of the yarns in the same warp column having the same movement in the weave plane. The yarns used here may in particular be carbon fiber yarns or ceramic fiber yarns such as silicon carbide (SiC) fibers, the invention not being limited to these types of yarns alone.

(8) FIG. 2 illustrates a loom **100** in accordance with one embodiment of the invention. The loom **100** is equipped with a Jacquard mechanism **101** supported by a superstructure not represented in FIG. 3. The loom **100** also comprises a harness **110** consisting of a tying-up board **111** and control yarns or heddles **113**, each heddle **113** being connected at one end to a control hook **102** of the Jacquard mechanism **101** and at the other end to one of the return springs **105** fixed to the frame **103** of the loom **100**. Each heddle **113** comprises an eyelet **114** crossed by a warp yarn **203**. The warp yarns **201** are organized at the level of the harness **110** of the loom in a plurality of layers and columns which are, as explained below, manipulated by the loom in order to allow the insertion of weft yarns **204** according to the weaving pattern(s) programmed in the loom. The warp yarns **203** extend in a horizontal direction D.sub.H corresponding to their direction of advance during the weaving. The weft yarns **204** are inserted between the warp yarns per column in a transverse direction DT perpendicular to the horizontal direction D.sub.H. In order to allow the introduction of each column of weft yarns during the weaving of the fibrous structure, a warp yarn take-up system (not represented in FIG. 2) is associated with the loom. This system, placed downstream of the loom, has the role of holding all of the warp yarns together in a clamping device and of allowing the warp yarns to advance by a determined distance after the insertion of each weft column.

(9) The heddles **113** and their associated eyelet **114** extend in an area Z in which the heddles **113** and the eyelets **114** are driven by a substantially vertical oscillating movement represented by the double arrow F. during the creation of a shed, as illustrated in FIG. 2, a portion of the heddles **113** is subjected to tensile forces exerted by the control hooks **102**. In this configuration, the heddles **113** make it possible to lift some warp yarns **203** and thus create a shed **104** allowing the passage of a rapier **120** for the introduction of weft yarns **204**.

(10) The rapier **120**, present downstream of the heddles **113**, is composed of a rod **121**, a first end of which is connected to an actuation system (not represented in FIG. 2) making it possible to reciprocate the rod **121** in the double direction D.sub.121. The other end of the rod **121** is provided with a clamp **122** which, after having passed through the shed **104** during the forward path of the rod **121**, grips a weft yarn **204** stored on a bobbin **130** to unwind it in the shed **104** during the return path of the rod **121**. The weft yarn **204** thus placed inside the shed **104** is cut in the vicinity of the bobbin **130** by a knife **140** and released at its other end by the clamp **122**.

(11) A reed **150** present upstream of the rapier **120** in its rest position is then folded down in order to pack the weft yarn(s) introduced into the shed **104** against the fell **205** of a fibrous structure **200**. The rapier **120** is then ready to again grip a new weft yarn **204** from the bobbin **130** and place it either again in the shed **104** or in a different shed depending on the defined weaving. A fibrous structure **200** having a 3D weaving between the warp yarns **203** and the weft yarns **204** is thus gradually formed.

(12) In accordance with the invention, the direction of layering of the weft yarns in each weft column is adjusted with the reed **150**. Indeed, the angle with which the reed beats up the weft yarns

against the fell **205** of the fibrous structure **200** determines the direction of layering of the weft yarns in each column of weft yarns. For this purpose, in the present invention, the loom is equipped with a reed comprising at least two stationary portions forming an angle therebetween, the reed being orientable in order to set the beat-up angle thereof on the fell of the woven fibrous structure as a function of the layering direction to be obtained in each column of weft yarns. The beat-up direction of the reed is parallel to the horizontal direction D.sub.H. The vertical position of the reed relative to the fibrous structure is further adjustable in order to position one or more stationary portions of the reed facing the fell of the fibrous structure.

(13) In the example described here, the reed **150** successively comprises first, second and third stationary portions **151**, **152** and **153** forming an angle therebetween, the first and second stationary portions **151** and **152** forming an angle $\alpha_{\text{sub.1}}$ therebetween while the second and third stationary portions **152** and **153** form an angle $\alpha_{\text{sub.2}}$ therebetween (FIGS. 3 to 6).

(14) The reed **150** is mounted on the loom by means of a positioning mechanism **170** secured for example to the frame **103**. The mechanism **170** comprises an angular adjustment device **171** comprising an axis of rotation or ball joint **1710** to which the lower end of the stationary portion **151** of the reed is connected here.

(15) The loom according to the invention also comprises means for vertically translating the reed or the fell of the fibrous structure.

(16) Concerning the vertical translation of the reed, the positioning mechanism **170** further comprises a vertical translation device **172** on which the angular adjustment device **171** is mounted, the vertical translation device **172** being secured to the frame **103** of the loom. The device **172** makes it possible to move the reed **150** upwards or downwards relative to a horizontal reference plane P.sub.H. The horizontal reference plane P.sub.H corresponds to the middle portion of the fell **205** of the fibrous structure in the vertical direction D.sub.V.

(17) Concerning the vertical translation of the fell of the fibrous structure, the loom further comprises a holding device **160** comprising a lower jaw **161** and an upper jaw **162** each connected to an actuation means (not represented in FIG. 2) which is able to hold the woven structure **200** on the one hand, and to move the jaws **160** and **161** in the vertical direction D.sub.V on the other hand. The holding device **160** makes it possible to move the fell **205** of the woven structure in the vertical direction D.sub.V upwards or downwards relative to the horizontal reference plane P.sub.H.

(18) The positioning mechanism **170** is thus able to orient the reed about the axis of rotation **1710** so as to adjust an angular position of the first, second and third stationary portions **151**, **152** and **153** of the reed **150** relative to the horizontal reference plane P.sub.H and to move the reed or the fell of the fibrous structure in the vertical direction D.sub.V so as to move said reed upwards or downwards relative to the horizontal reference plane P.sub.H.

(19) The reed **150** and/or the positioning mechanism **160** are mounted on a beat-up mechanism (not represented in FIG. 2) making it possible to beat up the fell **205** of the fibrous structure in a beat-up direction D.sub.F parallel to the horizontal direction D.sub.H.

(20) Thus, during the weaving of each section of a fibrous structure, the angular and vertical positions of the reed relative to the fell of the fibrous structure are adjusted so that one or more stationary portions of the reed beat up the fell of the fibrous structure at one or more determined angles.

(21) Several examples of 3D weaving with the loom **100** according to the invention are now described.

(22) FIG. 3 shows the weaving of a section **10** of fibrous structure by 3D weaving between a plurality of layers of warp yarns **11** and a plurality of layers of weft yarns **12**. The weft yarns are woven in the section **10** in a plurality of columns C.sub.T12 spaced from each other in the horizontal direction D.sub.H. The weft yarns of each column of weft yarns C.sub.T12 are juxtaposed in the thickness of the section **10** of fibrous structure in a determined layering direction D.sub.S12 perpendicular to the horizontal reference plane P.sub.H.

(23) During the weaving of the section **10**, the reed **150** is oriented about its axis of rotation **1710** so that the second stationary portion **152** is perpendicular to the horizontal reference plane P.sub.H. The reed **150** or the fell **15** of the section **10** of fibrous structure is positioned in the vertical direction D.sub.V so that the second stationary portion **152** is facing the fell **15**. Thus, each time the reed **150** beats up the fell **15** in the beat-up direction D.sub.F, the weft yarns of the section **10** are juxtaposed in each column of weft yarns in the layering direction D.sub.S10 which is perpendicular to the horizontal direction D.sub.H.

(24) FIG. 4 shows the weaving of a section **20** of fibrous structure by 3D weaving between a plurality of layers of warp yarns **21** and a plurality of layers of weft yarns **22**. The weft yarns are woven in the section **20** in a plurality of columns C.sub.T22 spaced from each other in the horizontal direction D.sub.H. The weft yarns **22** of each column of weft yarns C.sub.T22 of the section **20** are juxtaposed against the fell **25** of the section **20** of fibrous structure in a layering direction D.sub.S22 not perpendicular to the horizontal reference plane P.sub.H.

(25) During the weaving of the section **20** of fibrous structure, the reed **150** is oriented about its axis of rotation **1710** so that the third stationary portion **153** forms an angle $\beta_{\text{sub.22}}$ with the horizontal reference plane P.sub.H which is greater than 90° . The reed **150** or the fell **25** is positioned in the vertical direction D.sub.V so that the third stationary portion **153** is facing the fell **25**. Thus, each time the reed **150** beats up in the beat-up direction D.sub.F, the weft yarns of the section **20** are juxtaposed in each column of weft yarns in the layering direction D.sub.S22 forming the angle $\beta_{\text{sub.22}}$ with the horizontal reference plane P.sub.H.

(26) FIG. 5 shows the weaving of a section **30** of fibrous structure by 3D weaving between a plurality of layers of warp yarns **31** and a plurality of layers of weft yarns **32**. The weft yarns are woven in the section **30** in a plurality of columns C.sub.T32 spaced from each other in the horizontal direction D.sub.H. The weft yarns **32** of each column of weft yarns C.sub.T32 of the section **30** are juxtaposed against the fell **35** of the section **30** of fibrous structure in a first layering direction D.sub.S320 not perpendicular to the horizontal reference plane P.sub.H and a second layering direction D.sub.S321 perpendicular to the horizontal reference plane P.sub.H.

(27) During the weaving of the section **30** of fibrous structure, the reed **150** is oriented about its axis of rotation **1710** so that the first stationary portion **151** forms an angle $\beta_{\text{sub.320}}$ with the horizontal reference plane P.sub.H which is greater than 90° and so that the second stationary portion **152** forms an angle of 90° with the horizontal reference plane P.sub.H. The reed **150** or the fell **35** is positioned in the vertical direction D.sub.V so that a section of the first stationary portion **151** is facing a lower portion of the fell **35** and so that a section of the second stationary portion **152** is facing an upper portion of the fell **35**. Thus, each time the reed **150** beats up in the beat-up direction D.sub.F, the weft yarns of the section **30** are juxtaposed in each column of weft yarns in the first layering direction D.sub.S320 forming the angle $\beta_{\text{sub.320}}$ with the horizontal reference plane P.sub.H and in the second layering direction D.sub.S321 perpendicular to the horizontal reference plane P.sub.H. A section of fibrous structure with columns of weft yarns having a broken shape is thus obtained.

(28) FIG. 6 shows the weaving of a section **40** of fibrous structure by 3D weaving between a plurality of layers of warp yarns **41** and a plurality of layers of weft yarns **42**. The weft yarns are woven in the section **40** in a plurality of columns C.sub.T42 spaced from each other in the horizontal direction D.sub.H. The weft yarns **42** of each column of weft yarns C.sub.T42 of the section **40** are juxtaposed against the fell **45** of the section **40** of fibrous structure in a first layering direction D.sub.S420 not perpendicular to the horizontal reference plane P.sub.H and a second layering direction D.sub.S421 perpendicular to the horizontal reference plane P.sub.H.

(29) During the weaving of the section **40** of fibrous structure, the reed **150** is oriented about its axis of rotation **1710** so that the second stationary portion **152** forms an angle $\beta_{\text{sub.420}}$ with the horizontal reference plane P.sub.H which is less than 90° and so that the third stationary portion **153** forms an angle of 90° with the horizontal reference plane P.sub.H. The reed **150** or the fell **45**

is positioned in the vertical direction D.sub.V so that a section of the second stationary portion **152** is facing a lower portion of the fell **45** and so that a section of the third stationary portion **153** is facing an upper portion of the fell **45**. Thus, each time the reed **150** beat up in the beat-up direction D.sub.F, the weft yarns of the section **40** are juxtaposed in each column of weft yarns in the first layering direction D.sub.S420 forming the angle β .sub.420 with the horizontal reference plane P.sub.H and in the second layering direction D.sub.S421 perpendicular to the horizontal reference plane P.sub.H. A section of fibrous structure with columns of weft yarns having a broken shape is thus obtained.

(30) A fibrous structure in accordance with the invention, i.e. including a section with inclined or broken weft columns as described above, can be used as fibrous reinforcement for a composite material part. For this purpose, after possible shaping of the fibrous structure, the fibrous structure or preform is densified in order to form a composite material part. The densification of the fibrous preform intended to form the fibrous reinforcement of the part to be manufactured consists in filling the porosity of the preform, in all or part of its volume, with the material constituting the matrix. This densification can be performed in a manner known per se using the liquid process (CVL) or the gas process (CVI), or the ceramic filler injection process (Slurry Cast) or the silicon alloy impregnation process (MI or RMI) or using a sequence of one or more of these processes.

(31) The liquid process consists in impregnating the preform with a liquid composition containing a precursor of the matrix material. The precursor is usually in the form of a polymer, such as a high-performance epoxy resin, optionally diluted in a solvent. The preform is placed in a mold that can be tightly closed with a housing having the shape of the final molded blade. Then, the mold is closed and the liquid matrix precursor (for example a resin) is injected into the entire housing to impregnate the entire fibrous portion of the preform.

(32) The transformation of the precursor into a matrix, namely its polymerization, is performed by heat treatment, generally by heating of the mold, after removal of any solvent and crosslinking of the polymer, the preform being always held in the mold having a shape corresponding to that of the part to be produced.

(33) In the case of the formation of a carbon or ceramic matrix, the heat treatment consists in pyrolyzing the precursor to transform the matrix into a carbon or ceramic matrix depending on the precursor used and the pyrolysis conditions. For example, liquid precursors of ceramics, in particular of SiC or SiCN, can be resins of the polycarbosilane (PCS) or polytitanocarbosilane (PTCS) or polysilazane (PSZ) type, while liquid precursors of carbon can be resins with relatively high coke content, such as phenolic resins. Several consecutive cycles, from impregnation to heat treatment, can be performed to achieve the desired degree of densification.

(34) In the case in particular of the formation of an organic matrix, the densification of the fibrous preform can be performed by the well-known transfer molding process called RTM (Resin Transfer Molding). In accordance with the RTM process, the fibrous preform is placed in a mold having the external shape of the part to be produced. A thermosetting resin is injected into the inner space of the mold which comprises the fibrous preform. A pressure gradient is generally established in this inner space between the place where the resin is injected and the discharge orifices of the latter in order to monitor and optimize the impregnation of the preform with the resin.

(35) The densification of the preform can also be performed by polymer impregnation and pyrolysis (PIP), or by impregnation of a slurry cast, containing for example SiC and organic binders, followed by infiltration with liquid silicon (Melt infiltration).

(36) The densification of the fibrous preform can also be performed, in a known manner, by gas process through chemical vapor infiltration (CVI) of the matrix. The fibrous preform corresponding to the fibrous reinforcement of the blade to be produced is placed in a furnace into which a reaction gas phase is admitted. The pressure and the temperature prevailing in the furnace and the composition of the gas phase are chosen so as to allow the diffusion of the gas phase within the porosity of the preform to form the matrix by deposition, at the core of the material in contact with

the fibers, of a solid material resulting from a decomposition of a constituent of the gas phase or from a reaction between several constituents, unlike the pressure and temperature conditions specific to CVD (Chemical Vapor Deposition) processes which exclusively lead to a deposition on the surface of the material.

(37) The formation of a SiC matrix can be obtained with methyltrichlorosilane (MTS) giving SiC by decomposition of the MTS while a carbon matrix can be obtained with hydrocarbon gases such as methane and/or propane giving carbon by cracking.

(38) A densification combining liquid process and gas process can also be used to facilitate the implementation, limit the costs and the manufacturing cycles while obtaining satisfactory characteristics for the intended use.

(39) The densification methods described above make it possible to produce, from the fibrous structure of the invention, mainly parts made of organic matrix composite material (OMC), carbon matrix composite material (C/C) and ceramic matrix composite material (CMC). The organic matrix composite material (OMC), carbon matrix composite material (C/C) and ceramic matrix composite material (CMC) replace the metal material parts in some portions of turbomachines. Their use contributes to optimizing the performance of the aircrafts, in particular by improving the efficiency of the turbomachine and decreasing the overall mass of the turbomachine, which significantly reduces harmful emissions to the environment (CO, CO₂, NO_x, etc.).

(40) After densification, a composite material part is obtained.

(41) The fibrous structure and its manufacturing method according to the present invention can in particular be used to produce turbine ring sectors, stiffeners, turbomachine vanes or blades.

Claims

1. A Jacquard type loom intended to produce a fibrous structure by three-dimensional weaving between a plurality of warp yarns and a plurality of weft yarns extending in a horizontal direction, the loom comprising a plurality of control yarns able to move the warp yarns in a vertical direction, at least one rapier present downstream of the control yarns able to pull a weft yarn and a reed movable between a rest position upstream of the rapier and a beat-up position of a fell of the fibrous structure in a beat-up direction parallel to the horizontal direction, wherein the reed comprises at least two stationary portions forming an angle therebetween and in that the reed is mounted on the loom by means of a positioning mechanism able to at least orient the reed about an axis of rotation so as to adjust an angular position of said at least two stationary portions of the reed relative to a horizontal reference plane, the loom further comprising vertical translation means able to move the reed or the fell of the fibrous structure in the vertical direction so as to move said reed or said fell upwards or downwards relative to the horizontal reference plane.

2. The loom according to claim 1, wherein the angle formed between said at least two stationary portions is comprised between 60° and 80°.

3. A method for three-dimensional weaving in a single piece of a fibrous structure between a plurality of layers of warp yarns and a plurality of layers of weft yarns, the weaving being performed by a loom according to claim 1, the warp yarns extending in a horizontal direction corresponding to the direction of advance of said warp yarns, the weft yarns extending in a transverse direction, the weft yarns being woven in a plurality of columns spaced from each other in the horizontal direction, each column of weft yarns being positioned by the reed of the loom against the fell of the fibrous structure, the weft yarns of each column of weft yarns being juxtaposed in the thickness of the fibrous structure in a determined layering direction, the method comprising the weaving of at least one section with inclined weft columns in the fibrous structure, the weft yarns of the column of weft yarns of said at least one section with inclined weft columns being juxtaposed in at least one first layering direction not perpendicular to a horizontal reference plane.

4. The method according to claim 3, wherein the weft yarns of the columns of weft yarns of said at least one section with inclined weft columns are juxtaposed in different first and second layering directions.
 5. The method according to claim 3, wherein said at least one section with inclined weft columns is interlinked by weaving to one or more adjacent sections of the fibrous structure, the weft yarns of the columns of weft yarns of the adjacent section(s) being juxtaposed in a layering direction perpendicular to the horizontal direction.
 6. The method according to claim 3, wherein the first layering direction in said at least one section with inclined weft columns forms an angle with the horizontal direction comprised between 60° and 80°.
 7. A method for manufacturing a composite material part comprising: weaving a fibrous structure in accordance with the weaving method according to claim 3, shaping the fibrous structure so as to obtain a fibrous preform, densifying the fibrous preform by a matrix.
 8. A fibrous structure obtained in accordance with the three-dimensional weaving method according to claim 3, the fibrous structure having a three-dimensional weaving between a plurality of layers of warp yarns and a plurality of layers of weft yarns, the warp yarns extending in a horizontal direction, the weft yarns extending in a transverse direction, the structure comprising a plurality of columns of weft yarns spaced from each other in the horizontal direction, the weft yarns of each column of weft yarns being juxtaposed in the thickness of the fibrous structure in a determined layering direction, the fibrous structure comprising at least one section with inclined weft columns, the weft yarns of the columns of weft yarns of said at least one section with inclined weft columns being juxtaposed in at least one first layering direction not perpendicular to the horizontal direction.
 9. The structure according to claim 8, wherein the weft yarns of the columns of weft yarns of said at least one section with inclined weft columns are juxtaposed in different first and second layering directions.
 10. The structure according to claim 8, wherein said at least one section with inclined weft columns is interlinked by weaving to one or more adjacent sections of the fibrous structure, the weft yarns of the columns of weft yarns of the adjacent section(s) being juxtaposed in a layering direction perpendicular to the horizontal direction.
 11. The structure according to claim 8, wherein the first layering direction in said at least one section with inclined weft columns forms an angle with the horizontal direction comprised between 60° and 80°.
 12. A composite material part comprising a fibrous reinforcement densified by a matrix, characterized in that the fibrous reinforcement comprises a fibrous structure according to claim 8.
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